

LAB #2 — MULTITHREADING & ROBOT

1

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AGENDA

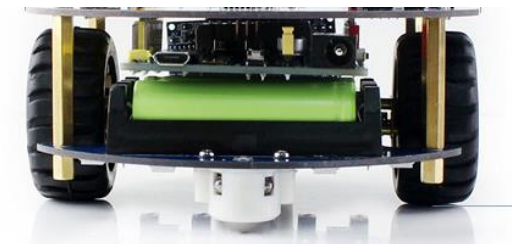
- Multithreading
- Getting Hands-On with AlphaBot2
- Connection to Alphabot2
- Testing Alphabot2
- Parallelizing Alphabot2 Tasks
- Implementation in Python
- Homework
- Words of Caution

MULTITHREADING

- Enables a program to run multiple threads concurrently.
- Allows tasks to execute independently and potentially in parallel to improve
 - performance
 - responsiveness

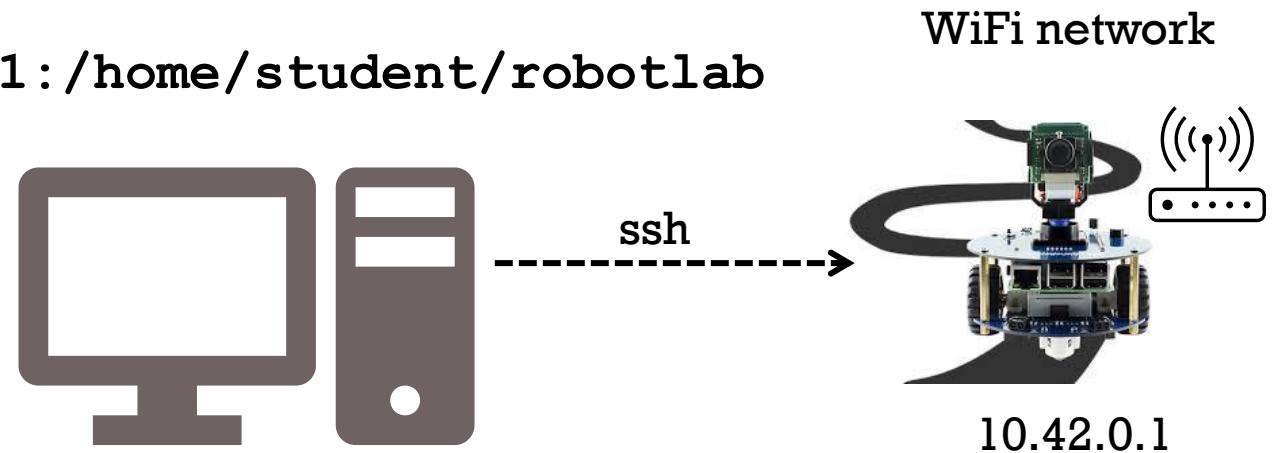
GETTING HANDS-ON WITH ALPHABOT2

- Raspberry Pi 3B
- Sensors
 - Infrared line tracking
 - Infrared obstacle avoidance
 - Camera
- Actuators
 - DC motors
 - Servo motors
 - RGB LED
 - Buzzer
- 2 chargeable batteries



CONNECTION TO ALPHABOT2

- Switch on the robot.
- Connect your computer or laptop to the AlphaBot2's WiFi network.
- `ssh student@10.42.0.1`
- Default password: `robotlab`
- VSCode
- `scp file.txt student@10.42.0.1:/home/student/robotlab`



TESTING ALPHABOT2

- **Run the provided program:**

```
python /home/student/robotlab/AlphaBot2.py
```

- Go through the program.
- Identify the different functions/tasks.
- Use CTRL-c to stop the program.



TESTING ALPHABOT2

- **Test the following tasks:**

- Black line tracking
- Object detection
- Buzzer
- RGB LEDs
- Camera

live stream <http://10.42.0.1:5000>

- **Report observations in 10 minutes**





TEST ALPHABOT2 TASKS

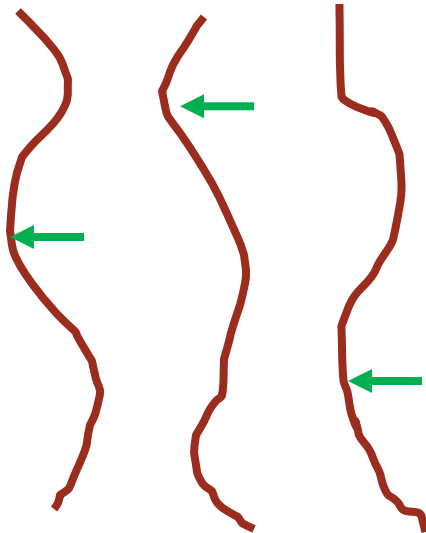
1. Do the sensors work?
2. Does Alphabot2 follow the line?
3. Does the camera work?
4. How would you rate the performance?

PARALLELIZING ALPHABOT2 TASKS

- **Single-thread**
- **Multiple threads**



single thread



multiple threads



IMPLEMENTATION IN PYTHON

```
import threading

def task():
    ...

# Create threads
t1 = threading.Thread(target=task)
t2 = threading.Thread(target=task)

# Start threads
t1.start()
t2.start()

# Wait for threads to finish
t1.join()
t2.join()
```

IMPLEMENTATION IN PYTHON

- Python's **Global Interpreter Lock (GIL)** can limit true parallelism for CPU-bound tasks.
- GIL is a mutex that allows **only one thread** to execute Python bytecode at a time, ***even on multi-core processors***.
- Implications:
 - I/O-bound can benefit from threading.
 - CPU-bound programs do not scale well with threading due to the GIL (use multiprocessing).



ECLECTX

HOMEWORK:

IMPROVE ALPHABOT2 PERFORMANCE

- Task 1: follow the black track continuously (no veering off the track)
- Task 2: detect objects
 - 1st object: it buzzes **once**
 - 2nd object: it buzzes **twice**
 - 3rd object: it buzzes **thrice**
- Task 3: recognize objects
 - shoe: sets the RGB LED to **red**,
 - bottle : sets the RGB LED to **yellow**
 - mug: sets the RGB LED to **green**
- Measure the time taken by the robot to complete one round trip on the track (using your mobile).

HOMework: REPORT AND DEMO

- In the lab session on 13.05.2026:
 - Submit your report and code on Moodle.
 - Demonstrate of all functionalities AlphaBot2.
- The grading scheme is provided on the Homework sheet #1.
- The exercise counts towards 25% of the overall grade.

WORDS OF CAUTION

- Before you switch off the robot, you need to shut down the Raspberry Pi using the command `sudo shutdown 0` (in 0 minutes).
- When recharging the AlphaBot2's batteries, be aware that the Raspberry Pi may lose power momentarily. This can cause a temporary loss of Wi-Fi connection, as the Pi may reboot or the Wi-Fi module may reset.
- Always ensure batteries are sufficiently charged before testing. Otherwise, the peripheral components may behave unpredictably.
- The distance at which an object is detected depends on the potentiometer settings, as described on the Waveshare website. Adjusting the potentiometer changes how far away an object can be before it is detected. However, adjusting the potentiometer can lead to issues—such as the green indicator LED staying on continuously, even when no object is present.

OPTIONAL: ALPHABOT2 RACE COMPETITION ON 13TH MAY 2026 @ 13:30 – 14:30

- The race track will be more complex and challenging.
- Each team will race their AlphaBot2 from start to finish, aiming for the **shortest time** while successfully completing tasks.
- Details about the tasks will be shared with you after the demo of Homework #1.
- The team with the **lowest overall time** wins!
- **Note:** Participation in the competition **will not affect your grade!**





HAVE FUN!

If you have any questions, please send us an email aca@eclectx.org.